Module 1: Excel Fundamentals

1. Introduction to Microsoft Excel

What is Microsoft Excel?

Microsoft Excel is a spreadsheet application developed by Microsoft. It is used to store, organize, calculate, analyze, and present data in rows and columns.

Example: A teacher can use Excel to maintain student names, attendance, marks, fees, and results.

Key features of Excel

- **Data organization:** Store information in a structured format.
- **Calculations:** Perform mathematical calculations.
- **Formulas and functions:** Calculate totals, averages, percentages, and more.
- **Data analysis:** Sort, filter, and analyze large datasets.
- **Charts and graphs:** Represent data visually.
- **Formatting:** Apply colors, borders, fonts, and number formats.
- **Automation:** Automate repetitive tasks using features such as macros.

Applications of Excel

Field	Practical use
Education	Student marksheets and attendance
Business	Sales and expense reports
Accounting	Budgets and financial records
HR	Employee records and salary sheets
Data analysis	Data cleaning and reporting
Inventory	Stock and product management

Practical activity

Create a student record sheet with these columns:

Roll No.	Student Name	Course	Fees
101	Rahul Patel	Excel	5,000
102	Priya Shah	Python	7,000
103	Amit Kumar	Web Design	6,000

Student task: Enter the data, make the headings bold, apply borders, and adjust the column widths.

2. Excel Interface and Workbook Management

The Excel interface contains different tools and sections that help users work with spreadsheets efficiently.

Main components of the Excel interface

1. Title Bar: Displays the name of the current workbook.
2. Quick Access Toolbar: Provides shortcuts for commands such as Save and Undo.
3. Ribbon: Contains Excel tools organized into tabs.
4. Ribbon Tabs: Home, Insert, Page Layout, Formulas, Data, Review, View, and others.
5. Name Box: Displays the address of the selected cell, such as A1.
6. Formula Bar: Displays or allows editing of a cell's content or formula.
7. Worksheet Area: The main grid where data is entered.
8. Row Headings: Numbered labels on the left side.
9. Column Headings: Lettered labels at the top.
10. Sheet Tabs: Used to switch between worksheets.
11. Status Bar: Displays information such as selection statistics and view controls.
12. Zoom Control: Changes the worksheet's display magnification.

Workbook vs. Worksheet

- Workbook: The entire Excel file.
- Worksheet: An individual spreadsheet inside the workbook.

For example, a workbook named Institute_Record.xlsx can contain three worksheets: Students, Attendance, and Fees.

Basic workbook operations

- Create a new workbook using Ctrl + N.
- Open an existing workbook using Ctrl + O.
- Save a workbook using Ctrl + S.

- Add a worksheet using the + button beside the sheet tabs.
- Rename a worksheet by double-clicking its sheet tab.
- Delete a worksheet by right-clicking its tab and selecting Delete.
- Move between worksheets by clicking their tabs.

Practical activity

Create a workbook called Institute_Management.xlsx and create three worksheets:

- Student Records
- Attendance
- Fees Records

Enter sample information in each worksheet and save the workbook.

3. Rows, Columns, Cells and Cell References

Excel organizes information using a grid structure.

Rows

Rows run horizontally across the worksheet and are identified by numbers: 1, 2, 3, 4, and so on.

Columns

Columns run vertically and are identified by letters: A, B, C, D, and so on.

Cells

A cell is the intersection of a row and a column. It is the basic unit used to store data.

For example, the intersection of column B and row 3 is cell B3.

Cell address

A cell address consists of a column letter followed by a row number.

Examples:

- A1 — Column A, Row 1
- C5 — Column C, Row 5
- D10 — Column D, Row 10

Cell references

Cell references are used in formulas to identify the cells containing the required data.

Reference type	Example	Explanation
Relative	A1	Changes when a formula is copied
Absolute	A1	Keeps the row and column fixed
Mixed	$A1	Keeps the column fixed
Mixed	A$1	Keeps the row fixed

For beginners, focus first on understanding relative cell references.

Practical example

Enter the following information in Excel:

Cell	Value
A1	Student Name
B1	Marks
A2	Rahul
B2	85
A3	Priya
B3	90

Click cell B2 and observe the Name Box. It should display B2.

Practical activity

Ask students to:

1. Select cell C5.
2. Enter the number 100.
3. Identify its row and column.
4. Select cell D10 and enter their name.
5. Identify the addresses of five different cells.

4. Data Entry and Editing

Data entry means entering information into Excel cells. Editing means changing, correcting, or updating existing information.

Types of data in Excel

- Text: Student names, city names, product names.
- Numbers: Marks, quantity, salary, prices.

- Dates: 29-10-2026.
- Times: 10:30 AM.
- Formulas: =A1+B1

How to enter data

1. Open Microsoft Excel.
2. Select a cell, such as A1.
3. Type the required information.
4. Press Enter to confirm the entry.
5. Continue entering data into other cells.

How to edit data

- Double-click a cell to edit its content.
- Select a cell and press F2 to edit it.
- Select a cell and type new content to replace the existing content.
- Use the Formula Bar to modify cell contents.
- Press Delete to clear selected cell contents.

Useful keyboard shortcuts

Shortcut	**Action**
Enter	Confirm entry
Tab	Move to the next cell
F2	Edit the active cell
Delete	Clear cell contents
Ctrl + Z	Undo the last action
Ctrl + Y	Redo an action
Ctrl + C	Copy
Ctrl + V	Paste
Ctrl + X	Cut
Ctrl + A	Select the current data region or all content, depending on context

Practical activity: Create a marksheet

Enter this data into Excel:

Student Name	English	Computer	Maths
Rahul	75	85	80
Priya	90	95	88
Amit	65	70	72
Neha	82	89	91

Tasks:

1. Enter all student records.
2. Correct one student's marks.
3. Add a new student.
4. Copy and paste a cell.
5. Undo and redo an action.
6. Clear the contents of one cell without deleting the entire row.

5. Saving and Managing Excel Files

Saving a workbook protects your work and allows you to reopen it later.

Common Excel file formats

Extension	Purpose
.xlsx	Standard Excel workbook
.xls	Older Excel workbook format
.xlsm	Excel workbook that supports macros
.csv	Plain-text, comma-separated data
.pdf	Sharing or printing a fixed-layout document

Save a new workbook

1. Open Excel.
2. Create or enter some data.
3. Press Ctrl + S.
4. Select a folder where you want to save the file.
5. Enter a filename, such as Student_Record.
6. Select the required file format.
7. Click Save.

Save As

The Save As option lets you save a workbook with a different name, in another location, or in a different format.

For example, save Student_Record.xlsx as Student_Record_Final.xlsx.

File management best practices

- Use meaningful filenames.
- Save files in the correct folders.
- Save your work regularly.
- Keep backup copies of important files.
- Use .xlsx for normal Excel workbooks.
- Use .csv when a system specifically requires CSV data.
- Avoid overwriting important original files without keeping a backup.

Practical activity

Create a folder named Excel_Practice. Inside it:

1. Create a workbook named Student_Record.xlsx.
2. Enter student information.
3. Save the workbook.
4. Use Save As to create Student_Record_Final.xlsx.
5. Close Excel and reopen the final file.
6. Verify that all the data is saved correctly.

Module 1: Homework and Assessment

Practical homework

- Write five uses of Microsoft Excel.
- Draw or identify eight components of the Excel interface.
- Explain the difference between a workbook and a worksheet.
- Create a student record containing at least 10 students.
- Practice entering, editing, copying, and deleting cell contents.
- Save the workbook using two different filenames.
- Explain relative, absolute, and mixed cell references with examples.